Homework 7 Solutions

Based on Lecture 7: Trigonometric Transformations, Equations, and Identities.

Angles without a degree symbol are in radians, and $k \in \mathbb{Z}$ means that k is any integer.

Section 1: Graph Transformations and Periods

Exercise 1. For $y = 3\sin(2x) - 1$ and $y = \frac{1}{2}\cos(x/3) + 2$, state the amplitude, period, midline, and range. Describe the horizontal and vertical scale factors relative to the parent graph.

Solution. Compare each function with $y = A\sin(B(x - C)) + D$ or $y = A\cos(B(x - C)) + D$.

For $y = 3\sin(2x) - 1$: $A = 3$, $B = 2$, $C = 0$, $D = -1$.

- Amplitude: $|A| = \boxed{3}$; period: $T = 2\pi/2 = \boxed{\pi}$.
- Midline: $\boxed{y = -1}$; range: $[-1 - 3,\ -1 + 3] = \boxed{[-4, 2]}$.
- Horizontal scale factor $1/|B| = \boxed{1/2}$: the graph of $\sin x$ is compressed horizontally toward the y-axis by a factor $1/2$. Vertical scale factor $\boxed{3}$: it is stretched vertically by a factor 3, then shifted down 1.

For $y = \frac{1}{2}\cos(x/3) + 2$: $A = 1/2$, $B = 1/3$, $C = 0$, $D = 2$.

- Amplitude: $\boxed{1/2}$; period: $T = 2\pi/(1/3) = \boxed{6\pi}$.
- Midline: $\boxed{y = 2}$; range: $[2 - \frac{1}{2},\ 2 + \frac{1}{2}] = \boxed{[3/2,\ 5/2]}$.
- Horizontal scale factor $1/|B| = \boxed{3}$: the graph of $\cos x$ is stretched horizontally away from the y-axis by a factor 3. Vertical scale factor $\boxed{1/2}$: it is compressed vertically by a factor $1/2$, then shifted up 2.

Check: $3\sin(2 \cdot \pi/4) - 1 = 2$ and $3\sin(2 \cdot 3\pi/4) - 1 = -4$ are the extreme values of the first function; $\frac{1}{2}\cos 0 + 2 = 5/2$ and $\frac{1}{2}\cos(3\pi/3) + 2 = 3/2$ (at $x = 3\pi$) are those of the second.

Exercise 2. Factor the input of $y = 2\sin(3x - 6) + 1$. State the horizontal shift and period, then explain why the shift is not 6 units to the right.

Solution. Factor 3 out of the entire input:

$$y = 2\sin(3x - 6) + 1 = \boxed{2\sin(3(x - 2)) + 1}.$$

So $B = 3$ and $C = 2$:

$$\text{horizontal shift} = \boxed{\text{2 units to the right}}, \qquad \text{period} = \frac{2\pi}{3} \implies \boxed{T = \frac{2\pi}{3}}.$$

The shift is the number subtracted from x itself, not from $3x$. A sine cycle starts where its input is 0, and $3x - 6 = 0$ at $x = 2$, not at $x = 6$ (at $x = 6$ the input is 12). The point rule $(u, v) \mapsto (2 + u/3,\ 1 + 2v)$ shows the order: first compress by $1/3$, then shift right 2; the parent starting point $(0, 0)$ moves to $(2, 1)$.

Shifting $y = 2\sin(3x) + 1$ right by 6 would replace x by $x - 6$, giving $2\sin(3(x - 6)) + 1 = 2\sin(3x - 18) + 1$, a different function. The number 6 is a shift only if it is applied to $y = \sin x$ *before* the compression ($\sin x \to \sin(x - 6) \to \sin(3x - 6)$); the compression by $1/3$ then shrinks that shift to $6/3 = 2$.

Check: $x = 2$ gives $2\sin 0 + 1 = 1$ (start of a cycle on the midline), and $x = 2 + \pi/6$ gives $2\sin(\pi/2) + 1 = 3$ (the maximum).

Exercise 3. Write the point transformation from $y = \cos x$ to $y = -2\cos(3(x + 1)) + 4$. Use it to move the parent points $(0, 1)$ and $(\pi, -1)$.

Solution. The input is already factored, with $x+1 = x-(-1)$. Thus $A=-2$, $B=3$, $C=-1$, $D=4$, and the rule $(u,v) \mapsto (C+u/B,\ D+Av)$ becomes

$$\boxed{(u,v) \longmapsto \left(-1+\frac{u}{3},\ 4-2v\right)}.$$

Then

$$(0,1) \longmapsto (-1+0,\ 4-2) = \boxed{(-1,2)}, \qquad (\pi,-1) \longmapsto \left(-1+\frac{\pi}{3},\ 4+2\right) = \boxed{\left(-1+\frac{\pi}{3},\ 6\right)}.$$

Check: at $x=-1$, $-2\cos 0+4=2$; at $x=-1+\pi/3$, $-2\cos(3\cdot\pi/3)+4=-2(-1)+4=6$. Because $A<0$, the parent maximum $(0,1)$ becomes the minimum value 2, and the parent minimum $(\pi,-1)$ becomes the maximum value 6.

Exercise 4. Sketch one cycle of $y=3\sin(2(x+1))-2$. Mark five key points and the midline, and state the maximum and minimum.

Solution. Here $A=3$, $B=2$, $C=-1$, $D=-2$. The amplitude is 3, the period is $T=2\pi/2=\pi$, the midline is $y=-2$, and the key points are spaced $T/4=\pi/4$ apart starting at $x=C=-1$. The point rule $(u,v)\mapsto(-1+u/2,\ -2+3v)$ moves the parent sine points $(0,0),(\pi/2,1),(\pi,0),(3\pi/2,-1),(2\pi,0)$ to

x	-1	$-1+\frac{\pi}{4}$	$-1+\frac{\pi}{2}$	$-1+\frac{3\pi}{4}$	$-1+\pi$
y	-2	1	-2	-5	-2

Therefore

$$\text{maximum} = -2+3 = \boxed{1} \text{ at } x=-1+\frac{\pi}{4}, \qquad \text{minimum} = -2-3 = \boxed{-5} \text{ at } x=-1+\frac{3\pi}{4},$$

and these repeat every π. The range is $[-5,1]$. Check: at $x=-1+\pi/4$ the input is $2(x+1)=\pi/2$, so $y=3(1)-2=1$; at $x=-1+3\pi/4$ it is $3\pi/2$, so $y=3(-1)-2=-5$. The cycle runs from $x=-1$ to $x=-1+\pi\approx 2.14$.

Exercise 5. Sketch one cycle of $y=-2\cos((x-2)/2)+1$. Explain how the negative sign changes the starting point of a cosine cycle, and state the period and range.

Solution. Factor the input: $(x-2)/2=\frac{1}{2}(x-2)$. So $A=-2$, $B=1/2$, $C=2$, $D=1$, and

$$\text{period} = \frac{2\pi}{1/2} = \boxed{4\pi}, \qquad \text{midline } y=1, \qquad \text{range} = [1-2,\ 1+2] = \boxed{[-1,3]}.$$

The point rule is $(u,v)\mapsto(2+2u,\ 1-2v)$, and the key points are spaced $T/4=\pi$ apart. The parent cosine points $(0,1),(\pi/2,0),(\pi,-1),(3\pi/2,0),(2\pi,1)$ move to

$$(2,-1), \quad (2+\pi,1), \quad (2+2\pi,3), \quad (2+3\pi,1), \quad (2+4\pi,-1).$$

A cosine cycle starts where its input is 0; here $\frac{1}{2}(x-2)=0$ at $x=2$. With a positive coefficient, $y=2\cos((x-2)/2)+1$, the cycle would start at its maximum $(2,3)$. The negative sign reflects the graph across the midline (the reflection happens before the vertical shift), so the parent value $v=1$ becomes $1-2\cdot 1=-1$: the cycle **starts at its minimum** $(2,-1)$, reaches its maximum 3 in the middle at $x=2+2\pi$, and ends at the minimum $(2+4\pi,-1)$. The starting x-value 2 does not change.

Check: $x=2$ gives $-2\cos 0+1=-1$; $x=2+\pi$ gives $-2\cos(\pi/2)+1=1$; $x=2+2\pi$ gives $-2\cos\pi+1=3$.

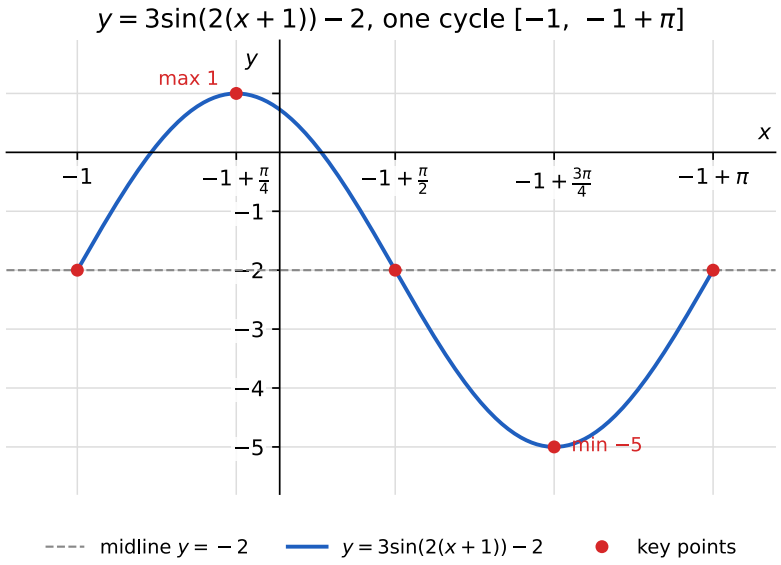


Figure 1: One cycle of y equals 3 sin(2(x plus 1)) minus 2 from x equals minus 1 to minus 1 plus pi, with midline y equals minus 2, maximum 1 at minus 1 plus pi over 4, and minimum minus 5 at minus 1 plus 3 pi over 4

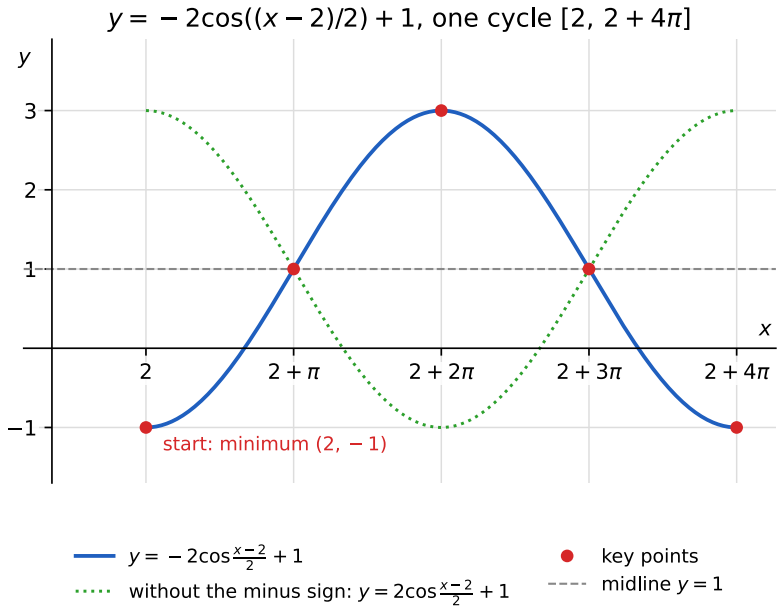


Figure 2: One cycle of y equals minus 2 cos((x minus 2) over 2) plus 1 from x equals 2 to 2 plus 4 pi, starting at the minimum (2, minus 1) with maximum 3 at 2 plus 2 pi, compared with the dotted unreflected curve that starts at its maximum

Section 2: Solving Trigonometric Equations

Exercise 6. Find all real solutions of $2\sin(x-1)-\sqrt{3}=0$. First write the solutions for the inner angle $u=x-1$.

Solution. Isolate the sine: $\sin(x-1)=\sqrt{3}/2$. With $u=x-1$, the standard equation $\sin u=\sqrt{3}/2=\sin(\pi/3)$ has all solutions

$$u=\frac{\pi}{3}+2k\pi \quad \text{or} \quad u=\pi-\frac{\pi}{3}+2k\pi=\frac{2\pi}{3}+2k\pi.$$

Substitute $u=x-1$ and add 1:

$$\boxed{x=1+\frac{\pi}{3}+2k\pi \quad \text{or} \quad x=1+\frac{2\pi}{3}+2k\pi}.$$

No division was needed, so the families keep the period 2π. Check: $x=1+\pi/3$ gives $2\sin(\pi/3)-\sqrt{3}=\sqrt{3}-\sqrt{3}=0$, and $x=1+2\pi/3$ gives $2\sin(2\pi/3)-\sqrt{3}=0$; adding $2k\pi$ does not change the sine.

Exercise 7. Find all real solutions of $\cos(3x-\pi/3)=0$. Combine any overlapping solution families into one family if possible.

Solution. Let $u=3x-\pi/3$. Since $\cos u=0=\cos(\pi/2)$,

$$u=\frac{\pi}{2}+2k\pi \quad \text{or} \quad u=-\frac{\pi}{2}+2k\pi.$$

The second family is $-\pi/2+2k\pi=\pi/2+(2k-1)\pi$, so it adds the odd multiples of π to $\pi/2$, while the first adds the even multiples. Together they form the single family

$$u=\frac{\pi}{2}+k\pi,$$

the top and bottom points $(0,\pm1)$ of the unit circle, half a turn apart. Substitute back, add $\pi/3$, and divide every term by 3:

$$3x-\frac{\pi}{3}=\frac{\pi}{2}+k\pi \quad\Longrightarrow\quad 3x=\frac{5\pi}{6}+k\pi \quad\Longrightarrow\quad \boxed{x=\frac{5\pi}{18}+\frac{k\pi}{3}}.$$

Kept separately, the two families are $x=5\pi/18+2k\pi/3$ and $x=-\pi/18+2k\pi/3$; since $5\pi/18-\pi/3=-\pi/18$, they are the even and odd members of the combined family. Check: $x=5\pi/18$ gives $3x-\pi/3=15\pi/18-6\pi/18=\pi/2$, and $x=-\pi/18$ gives $-\pi/6-\pi/3=-\pi/2$; cosine is 0 at both.

Exercise 8. Find all real solutions of $\sin(1-2x)=1/2$. Explain why a term $-k\pi$ can be rewritten as $k\pi$ when describing all solutions.

Solution. Let $u=1-2x$. Since $\sin u=1/2=\sin(\pi/6)$,

$$u=\frac{\pi}{6}+2k\pi \quad \text{or} \quad u=\frac{5\pi}{6}+2k\pi.$$

Substitute back, subtract 1, and divide every term by -2:

$$1-2x=\frac{\pi}{6}+2k\pi \implies -2x=-1+\frac{\pi}{6}+2k\pi \implies x=\frac{1}{2}-\frac{\pi}{12}-k\pi,$$

$$1-2x=\frac{5\pi}{6}+2k\pi \implies -2x=-1+\frac{5\pi}{6}+2k\pi \implies x=\frac{1}{2}-\frac{5\pi}{12}-k\pi.$$

As k runs through all integers, so does $-k$: the map $k\mapsto -k$ just lists $\ldots,2,1,0,-1,-2,\ldots$ in the opposite order. Hence the sets $\{-k\pi : k\in\mathbb{Z}\}$ and $\{k\pi : k\in\mathbb{Z}\}$ are the same set $\{\ldots,-2\pi,-\pi,0,\pi,2\pi,\ldots\}$, and renaming $-k$ as k gives

$$\boxed{x=\frac{1}{2}-\frac{\pi}{12}+k\pi \quad \text{or} \quad x=\frac{1}{2}-\frac{5\pi}{12}+k\pi}.$$

Check: $x = \frac{1}{2} - \frac{\pi}{12}$ gives $1 - 2x = \pi/6$, and $x = \frac{1}{2} - \frac{5\pi}{12} + \pi$ gives $1 - 2x = 5\pi/6 - 2\pi$; the sine is $1/2$ in both cases.

Exercise 9. Find all real solutions of $\tan(2x - \pi/6) = -\sqrt{3}$. Check the original domain.

Solution. Let $u = 2x - \pi/6$. Since $\tan u = -\sqrt{3} = \tan(-\pi/3)$ and tangent has period π,

$$u = -\frac{\pi}{3} + k\pi.$$

Substitute back, add $\pi/6$, and divide every term by 2:

$$2x - \frac{\pi}{6} = -\frac{\pi}{3} + k\pi \quad \Longrightarrow \quad 2x = -\frac{\pi}{6} + k\pi \quad \Longrightarrow \quad \boxed{x = -\frac{\pi}{12} + \frac{k\pi}{2}}$$

(equivalently $x = 5\pi/12 + k\pi/2$).

Domain: the original expression requires $\cos(2x - \pi/6) \neq 0$, that is, $2x - \pi/6 \neq \pi/2 + m\pi$, or $x \neq \pi/3 + m\pi/2$ for $m \in \mathbb{Z}$. Every candidate has inner angle $-\pi/3 + k\pi$, where $\cos(-\pi/3 + k\pi) = \pm\frac{1}{2} \neq 0$. So every candidate is in the domain and none is discarded. Check: $x = -\pi/12$ gives $\tan(-\pi/3) = -\sqrt{3}$, and $x = 5\pi/12$ gives $\tan(5\pi/6 - \pi/6) = \tan(2\pi/3) = -\sqrt{3}$.

Exercise 10. Solve $2\sin^2 x = \sin x$ for all real x by factoring, then list the solutions on $[0, 2\pi)$. Which solutions would be lost by dividing by $\sin x$?

Solution. Move everything to one side and factor:

$$2\sin^2 x - \sin x = 0 \quad \Longrightarrow \quad \sin x\,(2\sin x - 1) = 0.$$

Either $\sin x = 0$, giving $x = k\pi$, or $\sin x = 1/2$, giving $x = \pi/6 + 2k\pi$ or $x = 5\pi/6 + 2k\pi$. All real solutions:

$$\boxed{x = k\pi \quad \text{or} \quad x = \frac{\pi}{6} + 2k\pi \quad \text{or} \quad x = \frac{5\pi}{6} + 2k\pi}.$$

On $[0, 2\pi)$, the family $k\pi$ contributes 0 and π (2π is excluded), and the other two contribute $\pi/6$ and $5\pi/6$:

$$\boxed{x = 0,\ \frac{\pi}{6},\ \frac{5\pi}{6},\ \pi}.$$

Dividing by $\sin x$ is allowed only where $\sin x \neq 0$; it leaves $2\sin x = 1$ and throws away the whole factor $\sin x = 0$. It would lose $\boxed{x = k\pi}$, which on $[0, 2\pi)$ are $x = 0$ and $x = \pi$. None of them is recovered from $2\sin x = 1$, since $\sin(k\pi) = 0 \neq \frac{1}{2}$. Check: $x = \pi/6$ gives $2(1/2)^2 = 1/2 = \sin(\pi/6)$; at $x = 0$ and $x = \pi$ both sides are 0.

Section 3: Proving the Cosine Sum Formula

Exercise 11. In the proof, why did we choose Q at angle $-b$ rather than angle b? If we choose $Q = (\cos b, \sin b)$ instead and rotate by $-b$, what formula does the same distance argument give?

Solution. The goal is $\cos(a + b)$. The angle from Q (at $-b$) to P (at a) is $a - (-b) = a + b$. Rotating by b moves Q to angle 0, which is the point $(1, 0)$, and moves P to angle $a + b$. With one endpoint at $(1, 0)$, the squared chord length is simply $2 - 2\cos(a + b)$, so the rotated chord measures exactly $\cos(a + b)$. If Q were at angle b, the angle between Q and P would be $a - b$, and bringing Q to $(1, 0)$ requires rotating by $-b$, which sends P to angle $a - b$. That produces the difference formula instead.

Carry it out with $P = (\cos a, \sin a)$ and $Q = (\cos b, \sin b)$:

$$\begin{aligned} PQ^2 &= (\cos a - \cos b)^2 + (\sin a - \sin b)^2 \\ &= (\cos^2 a + \sin^2 a) + (\cos^2 b + \sin^2 b) - 2\cos a \cos b - 2\sin a \sin b \\ &= 2 - 2\cos a \cos b - 2\sin a \sin b. \end{aligned}$$

Rotate both points through angle $-b$. Then Q moves to angle 0 and P to angle $a-b$:

$$P' = (\cos(a-b), \sin(a-b)), \qquad Q' = (1,0),$$

$$\begin{aligned} P'Q'^2 &= (\cos(a-b)-1)^2 + \sin^2(a-b) \\ &= \cos^2(a-b) + \sin^2(a-b) - 2\cos(a-b) + 1 \\ &= 2 - 2\cos(a-b). \end{aligned}$$

A rotation preserves distance, so

$$2 - 2\cos(a-b) = 2 - 2\cos a\cos b - 2\sin a\sin b.$$

Subtracting 2 and dividing by -2 gives the **cosine difference formula**

$$\boxed{\cos(a-b) = \cos a\cos b + \sin a\sin b}.$$

As in the lecture, the argument works for all real a, b. (Replacing b by $-b$ in it recovers the sum formula.)

Exercise 12. Use the cosine sum formula to find $\cos(7\pi/12)$ exactly. Check that its sign agrees with its quadrant.

Solution. Since $7\pi/12 = 4\pi/12 + 3\pi/12 = \pi/3 + \pi/4$,

$$\begin{aligned} \cos\frac{7\pi}{12} &= \cos\frac{\pi}{3}\cos\frac{\pi}{4} - \sin\frac{\pi}{3}\sin\frac{\pi}{4} \\ &= \frac{1}{2}\cdot\frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2}\cdot\frac{\sqrt{2}}{2} = \boxed{\frac{\sqrt{2}-\sqrt{6}}{4}} \approx -0.2588. \end{aligned}$$

Sign check: $\pi/2 = 6\pi/12 < 7\pi/12 < 12\pi/12 = \pi$, so $7\pi/12 = 105^\circ$ is in quadrant II, where cosine (the x-coordinate) is negative. Since $\sqrt{6} > \sqrt{2}$, the value $(\sqrt{2}-\sqrt{6})/4$ is indeed negative. Cross-check with Exercise 13: $\cos(\pi/12 + \pi/2) = -\sin(\pi/12) = -(\sqrt{6}-\sqrt{2})/4$, using the lecture's value of $\sin(\pi/12)$.

Exercise 13. Set $b = \pi/2$ in the cosine sum formula to show $\cos(a + \pi/2) = -\sin a$. Then set $b = \pi$ to show $\cos(a+\pi) = -\cos a$.

Solution. With $b = \pi/2$, use $\cos(\pi/2) = 0$ and $\sin(\pi/2) = 1$:

$$\cos\left(a + \frac{\pi}{2}\right) = \cos a\cos\frac{\pi}{2} - \sin a\sin\frac{\pi}{2} = \cos a\cdot 0 - \sin a\cdot 1 = \boxed{-\sin a}.$$

With $b = \pi$, use $\cos\pi = -1$ and $\sin\pi = 0$:

$$\cos(a+\pi) = \cos a\cos\pi - \sin a\sin\pi = \cos a\cdot(-1) - \sin a\cdot 0 = \boxed{-\cos a}.$$

Both hold for every real a, because the sum formula does. Geometrically, a quarter turn sends $(\cos a, \sin a)$ to $(-\sin a, \cos a)$ and a half turn sends it to $(-\cos a, -\sin a)$. Check with $a = \pi/3$: $\cos(5\pi/6) = -\sqrt{3}/2 = -\sin(\pi/3)$ and $\cos(4\pi/3) = -1/2 = -\cos(\pi/3)$.

Section 4: One Sum Formula Gives the Others

Exercise 14. Starting with the sine sum formula alone, use $\cos(a+b) = \sin((\pi/2 - a) - b)$ to derive the cosine sum formula. First obtain the sine difference formula by replacing b with $-b$.

Solution. Assume only

$$\sin(a+b) = \sin a\cos b + \cos a\sin b \qquad \text{for all real } a, b,$$

together with the facts that cosine is even, sine is odd, and the cofunction identities $\sin(\pi/2 - x) = \cos x$ and $\cos(\pi/2 - x) = \sin x$ (proved geometrically in Section 1 of the lecture, without any sum formula).

Step 1: sine difference. Replace b by $-b$ and use $\cos(-b)=\cos b$, $\sin(-b)=-\sin b$:

$$\sin(a-b)=\sin a\cos(-b)+\cos a\sin(-b)=\boxed{\sin a\cos b-\cos a\sin b}.$$

Step 2: cosine sum. By the cofunction identity with $x=a+b$,

$$\cos(a+b)=\sin\left(\frac{\pi}{2}-(a+b)\right)=\sin\left(\left(\frac{\pi}{2}-a\right)-b\right).$$

Apply Step 1 with first angle $\pi/2-a$ and second angle b, then the cofunction identities with $x=a$:

$$\begin{aligned}\cos(a+b)&=\sin\left(\frac{\pi}{2}-a\right)\cos b-\cos\left(\frac{\pi}{2}-a\right)\sin b\\&=\boxed{\cos a\cos b-\sin a\sin b}.\end{aligned}$$

Every step holds for all real a,b, so the cosine sum formula holds for all real a,b.

Exercise 15. Find exact values of $\sin(5\pi/12)$, $\cos(\pi/12)$, and $\tan(5\pi/12)$ using the sum or difference formulas.

Solution. Use $5\pi/12=3\pi/12+2\pi/12=\pi/4+\pi/6$ and $\pi/12=\pi/4-\pi/6$:

$$\sin\frac{5\pi}{12}=\sin\frac{\pi}{4}\cos\frac{\pi}{6}+\cos\frac{\pi}{4}\sin\frac{\pi}{6}=\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2}+\frac{\sqrt{2}}{2}\cdot\frac{1}{2}=\boxed{\frac{\sqrt{6}+\sqrt{2}}{4}}\approx 0.9659,$$

$$\cos\frac{\pi}{12}=\cos\frac{\pi}{4}\cos\frac{\pi}{6}+\sin\frac{\pi}{4}\sin\frac{\pi}{6}=\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2}+\frac{\sqrt{2}}{2}\cdot\frac{1}{2}=\boxed{\frac{\sqrt{6}+\sqrt{2}}{4}}\approx 0.9659.$$

For the tangent, $1-\tan\frac{\pi}{4}\tan\frac{\pi}{6}=1-\frac{1}{\sqrt{3}}\neq 0$, so the sum formula applies:

$$\tan\frac{5\pi}{12}=\frac{\tan\frac{\pi}{4}+\tan\frac{\pi}{6}}{1-\tan\frac{\pi}{4}\tan\frac{\pi}{6}}=\frac{1+1/\sqrt{3}}{1-1/\sqrt{3}}=\frac{\sqrt{3}+1}{\sqrt{3}-1}=\frac{(\sqrt{3}+1)^2}{3-1}=\frac{4+2\sqrt{3}}{2}=\boxed{2+\sqrt{3}}\approx 3.732.$$

Checks: $\sin(5\pi/12)=\cos(\pi/12)$ is expected, since $5\pi/12=\pi/2-\pi/12$. Also, with the lecture's $\cos(5\pi/12)=(\sqrt{6}-\sqrt{2})/4$,

$$\frac{\sin(5\pi/12)}{\cos(5\pi/12)}=\frac{\sqrt{6}+\sqrt{2}}{\sqrt{6}-\sqrt{2}}=\frac{(\sqrt{6}+\sqrt{2})^2}{6-2}=\frac{8+4\sqrt{3}}{4}=2+\sqrt{3},$$

and $(2+\sqrt{3})(2-\sqrt{3})=1$ matches $\tan(5\pi/12)=1/\tan(\pi/12)$.

Exercise 16. Suppose $\sin a=3/5$ with a in quadrant II, and $\cos b=12/13$ with b in quadrant IV. Find $\sin(a+b)$ and $\cos(a-b)$ exactly.

Solution. First find the missing values with $\sin^2\theta+\cos^2\theta=1$ and the quadrant signs:

$$\cos^2 a=1-\frac{9}{25}=\frac{16}{25},\quad \cos a<0 \text{ in QII } \Longrightarrow\ \cos a=-\frac{4}{5};$$

$$\sin^2 b=1-\frac{144}{169}=\frac{25}{169},\quad \sin b<0 \text{ in QIV } \Longrightarrow\ \sin b=-\frac{5}{13}.$$

Then

$$\sin(a+b)=\sin a\cos b+\cos a\sin b=\frac{3}{5}\cdot\frac{12}{13}+\left(-\frac{4}{5}\right)\left(-\frac{5}{13}\right)=\frac{36}{65}+\frac{20}{65}=\boxed{\frac{56}{65}},$$

$$\cos(a-b)=\cos a\cos b+\sin a\sin b=\left(-\frac{4}{5}\right)\frac{12}{13}+\frac{3}{5}\left(-\frac{5}{13}\right)=-\frac{48}{65}-\frac{15}{65}=\boxed{-\frac{63}{65}}.$$

Numerical check: $a=\pi-\arcsin(3/5)\approx 2.4981$ and $b=-\arccos(12/13)\approx -0.3948$, so $\sin(a+b)\approx\sin 2.1033\approx 0.8615$ and $\cos(a-b)\approx\cos 2.8929\approx -0.9692$, matching $56/65$ and $-63/65$.

Exercise 17. Simplify without a calculator:

$$\sin\frac{\pi}{7}\cos\frac{2\pi}{7}+\cos\frac{\pi}{7}\sin\frac{2\pi}{7}, \qquad \cos(x+y)\cos y+\sin(x+y)\sin y.$$

Solution. The first expression has the form $\sin a\cos b+\cos a\sin b$ with $a=\pi/7$ and $b=2\pi/7$, so by the sine sum formula

$$\sin\frac{\pi}{7}\cos\frac{2\pi}{7}+\cos\frac{\pi}{7}\sin\frac{2\pi}{7}=\sin\left(\frac{\pi}{7}+\frac{2\pi}{7}\right)=\boxed{\sin\frac{3\pi}{7}}.$$

(Equivalently $\cos(\pi/14)$, since $\pi/2-3\pi/7=\pi/14$.)

The second has the form $\cos A\cos B+\sin A\sin B$ with $A=x+y$ and $B=y$, so by the cosine difference formula

$$\cos(x+y)\cos y+\sin(x+y)\sin y=\cos((x+y)-y)=\boxed{\cos x}.$$

Check with $x=\pi/3$, $y=\pi/6$: $\cos\frac{\pi}{2}\cos\frac{\pi}{6}+\sin\frac{\pi}{2}\sin\frac{\pi}{6}=0+\frac{1}{2}=\cos\frac{\pi}{3}$.

Exercise 18 (sum-to-product formulas). Use the sine and cosine sum and difference formulas to prove all four identities:

$$\sin u+\sin v=2\sin\frac{u+v}{2}\cos\frac{u-v}{2},$$
$$\sin u-\sin v=2\cos\frac{u+v}{2}\sin\frac{u-v}{2},$$
$$\cos u+\cos v=2\cos\frac{u+v}{2}\cos\frac{u-v}{2},$$
$$\cos u-\cos v=-2\sin\frac{u+v}{2}\sin\frac{u-v}{2}.$$

Hint: set $a=(u+v)/2$ and $b=(u-v)/2$, so $u=a+b$ and $v=a-b$. Expand $\sin(a\pm b)$ and $\cos(a\pm b)$, then add or subtract the resulting expressions.

Solution. Let u,v be any real numbers and set

$$a=\frac{u+v}{2}, \qquad b=\frac{u-v}{2}.$$

Then $a+b=\frac{2u}{2}=u$ and $a-b=\frac{2v}{2}=v$. Since the sum and difference formulas hold for all real a,b,

$$\sin u=\sin(a+b)=\sin a\cos b+\cos a\sin b,$$
$$\sin v=\sin(a-b)=\sin a\cos b-\cos a\sin b,$$
$$\cos u=\cos(a+b)=\cos a\cos b-\sin a\sin b,$$
$$\cos v=\cos(a-b)=\cos a\cos b+\sin a\sin b.$$

Add or subtract the matching pairs. In each case two of the four products cancel and the other two double:

$$\sin u+\sin v=2\sin a\cos b=\boxed{2\sin\frac{u+v}{2}\cos\frac{u-v}{2}},$$
$$\sin u-\sin v=2\cos a\sin b=\boxed{2\cos\frac{u+v}{2}\sin\frac{u-v}{2}},$$
$$\cos u+\cos v=2\cos a\cos b=\boxed{2\cos\frac{u+v}{2}\cos\frac{u-v}{2}},$$
$$\cos u-\cos v=-2\sin a\sin b=\boxed{-2\sin\frac{u+v}{2}\sin\frac{u-v}{2}}.$$

Because u and v were arbitrary, all four identities hold for every real u,v. ■

Check with $u=\pi/2$, $v=\pi/6$ (so $a=\pi/3$, $b=\pi/6$): $\sin u+\sin v = 1+\frac{1}{2}=\frac{3}{2}$ and $2\sin\frac{\pi}{3}\cos\frac{\pi}{6} = 2\cdot\frac{\sqrt{3}}{2}\cdot\frac{\sqrt{3}}{2}=\frac{3}{2}$; $\cos u-\cos v = 0-\frac{\sqrt{3}}{2}$ and $-2\sin\frac{\pi}{3}\sin\frac{\pi}{6} = -2\cdot\frac{\sqrt{3}}{2}\cdot\frac{1}{2}=-\frac{\sqrt{3}}{2}$.

Exercise 19. Solve $\sin x+\cos x=1$ on $[0,2\pi)$ by rewriting the left side as $\sqrt{2}\sin(x+\pi/4)$.

Solution. First confirm the rewrite with the sine sum formula:

$$\sqrt{2}\sin\left(x+\frac{\pi}{4}\right)=\sqrt{2}\left(\sin x\cos\frac{\pi}{4}+\cos x\sin\frac{\pi}{4}\right)=\sqrt{2}\cdot\frac{\sqrt{2}}{2}(\sin x+\cos x)=\sin x+\cos x.$$

So the equation is $\sqrt{2}\sin(x+\pi/4)=1$, or $\sin(x+\pi/4)=1/\sqrt{2}=\sqrt{2}/2$. With $u=x+\pi/4$,

$$u=\frac{\pi}{4}+2k\pi \quad\text{or}\quad u=\frac{3\pi}{4}+2k\pi,$$

so the general solutions are $x=2k\pi$ or $x=\pi/2+2k\pi$. For $x\in[0,2\pi)$ the inner angle satisfies $\pi/4\le u<9\pi/4$, which contains $u=\pi/4$ and $u=3\pi/4$ only ($9\pi/4$ is excluded and $11\pi/4$ is too large). Hence

$$\boxed{x=0 \quad\text{or}\quad x=\frac{\pi}{2}}.$$

Check: $\sin 0+\cos 0=0+1=1$ and $\sin\frac{\pi}{2}+\cos\frac{\pi}{2}=1+0=1$. The graph of $y=\sqrt{2}\sin(x+\pi/4)$ has amplitude $\sqrt{2}$ and is the sine graph shifted left $\pi/4$; it meets $y=1$ at $x=0$ and $x=\pi/2$ (and again at 2π, which is excluded).

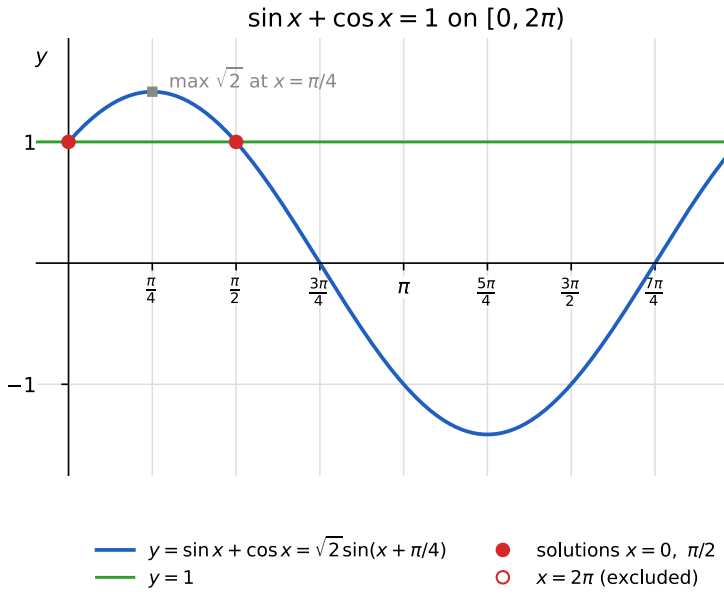


Figure 3: Graph of y equals sin x plus cos x, which equals root 2 sin(x plus pi over 4), on 0 to 2 pi, meeting the line y equals 1 at x equals 0 and x equals pi over 2, with the excluded endpoint 2 pi shown as an open circle

Section 5: Double-Angle Formulas

Exercise 20. Suppose $\sin\theta=-5/13$ and θ is in quadrant III. Find $\sin(2\theta)$, $\cos(2\theta)$, and $\tan(2\theta)$ exactly.

Solution. In quadrant III cosine is negative, so

$$\cos^2\theta=1-\frac{25}{169}=\frac{144}{169},\qquad \cos\theta=-\frac{12}{13}.$$

Then

$$\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(-\frac{5}{13}\right)\left(-\frac{12}{13}\right) = \boxed{\frac{120}{169}},$$

$$\cos(2\theta) = 1 - 2\sin^2\theta = 1 - 2\cdot\frac{25}{169} = \boxed{\frac{119}{169}}, \qquad \tan(2\theta) = \frac{120/169}{119/169} = \boxed{\frac{120}{119}}.$$

Checks: $\cos^2\theta - \sin^2\theta = \frac{144-25}{169} = \frac{119}{169}$; with $\tan\theta = 5/12$, $\frac{2\tan\theta}{1-\tan^2\theta} = \frac{5/6}{119/144} = \frac{120}{119}$; and $120^2 + 119^2 = 14400 + 14161 = 28561 = 169^2$. The signs are consistent: since $\tan\theta = 5/12 < 1$, we have $\pi < \theta < 5\pi/4$, so $2\pi < 2\theta < 5\pi/2$ and 2θ is coterminal with a quadrant I angle.

Exercise 21. Rewrite $\sin x\cos x$ and $\cos^4 x - \sin^4 x$ using double-angle identities.

Solution. Dividing $\sin(2x) = 2\sin x\cos x$ by 2,

$$\boxed{\sin x\cos x = \frac{1}{2}\sin(2x)}.$$

Factor the difference of squares and use $\cos^2 x + \sin^2 x = 1$:

$$\cos^4 x - \sin^4 x = (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x) = (\cos^2 x - \sin^2 x)\cdot 1 = \boxed{\cos(2x)}.$$

Both hold for every real x. Check at $x = \pi/6$: $\sin\frac{\pi}{6}\cos\frac{\pi}{6} = \frac{\sqrt{3}}{4} = \frac{1}{2}\sin\frac{\pi}{3}$, and $\cos^4\frac{\pi}{6} - \sin^4\frac{\pi}{6} = \frac{9}{16} - \frac{1}{16} = \frac{1}{2} = \cos\frac{\pi}{3}$.

Exercise 22. Find all real solutions of $\cos(2x) = \sin x$. Choose a form of $\cos(2x)$ that gives a quadratic in $\sin x$.

Solution. Use $\cos(2x) = 1 - 2\sin^2 x$, which involves only $\sin x$:

$$1 - 2\sin^2 x = \sin x \quad\Longrightarrow\quad 2\sin^2 x + \sin x - 1 = 0 \quad\Longrightarrow\quad (2\sin x - 1)(\sin x + 1) = 0.$$

(Expanding, $(2s-1)(s+1) = 2s^2 + s - 1$.)

- $\sin x = \frac{1}{2}$: $x = \dfrac{\pi}{6} + 2k\pi$ or $x = \dfrac{5\pi}{6} + 2k\pi$.
- $\sin x = -1$: $x = \dfrac{3\pi}{2} + 2k\pi$.

All real solutions:

$$\boxed{x = \frac{\pi}{6} + 2k\pi \quad\text{or}\quad x = \frac{5\pi}{6} + 2k\pi \quad\text{or}\quad x = \frac{3\pi}{2} + 2k\pi}.$$

These three families are spaced $2\pi/3$ apart ($\pi/6,\ \pi/6 + 2\pi/3 = 5\pi/6,\ \pi/6 + 4\pi/3 = 3\pi/2$), so they combine into the single family $x = \pi/6 + 2k\pi/3$. On $[0, 2\pi)$ they are $\pi/6,\ 5\pi/6,\ 3\pi/2$. Check: $\cos\frac{\pi}{3} = \frac{1}{2} = \sin\frac{\pi}{6}$; $\cos\frac{5\pi}{3} = \frac{1}{2} = \sin\frac{5\pi}{6}$; $\cos 3\pi = -1 = \sin\frac{3\pi}{2}$.

Exercise 23. Derive $\tan(2x) = 2\tan x/(1 - \tan^2 x)$ and state the restrictions needed for the displayed identity. Give an x for which $\tan(2x)$ exists but the expression on the right is undefined.

Solution. Where $\cos(2x) \neq 0$, the double-angle formulas give

$$\tan(2x) = \frac{\sin(2x)}{\cos(2x)} = \frac{2\sin x\cos x}{\cos^2 x - \sin^2 x}.$$

If also $\cos x \neq 0$, divide the numerator and denominator by $\cos^2 x$:

$$\tan(2x) = \frac{\dfrac{2\sin x\cos x}{\cos^2 x}}{\dfrac{\cos^2 x - \sin^2 x}{\cos^2 x}} = \frac{2\dfrac{\sin x}{\cos x}}{1 - \dfrac{\sin^2 x}{\cos^2 x}} = \boxed{\frac{2\tan x}{1 - \tan^2 x}}.$$

(This is also the tangent sum formula with $a = b = x$.)

Restrictions for the displayed right side:

- $\tan x$ must be defined: $\cos x \neq 0$, i.e. $x \neq \dfrac{\pi}{2} + k\pi$;
- the denominator must be nonzero: $\tan^2 x \neq 1$, i.e. $\tan x \neq \pm 1$, so $x \neq \dfrac{\pi}{4} + \dfrac{k\pi}{2}$.

So the identity holds for

$$\boxed{x \neq \frac{\pi}{2} + k\pi \quad \text{and} \quad x \neq \frac{\pi}{4} + \frac{k\pi}{2}}.$$

When $\cos x \neq 0$, the second condition is the same as $\cos(2x) \neq 0$, because $\cos(2x) = \cos^2 x\,(1 - \tan^2 x)$.

Example: $\boxed{x = \pi/2}$. Then $\tan(2x) = \tan \pi = 0$ exists, but $\tan(\pi/2)$ is undefined, so $2\tan x/(1 - \tan^2 x)$ is undefined. (Every $x = \pi/2 + k\pi$ works the same way, since $\tan(\pi + 2k\pi) = 0$.)

Exercise 24. Use $\cos(2x) = 1 - 2\sin^2 x$ to rewrite $\sin^2 x$ as a transformed cosine function. Use this form to state its fundamental period, midline, and range, and sketch one cycle.

Solution. Solve $\cos(2x) = 1 - 2\sin^2 x$ for $\sin^2 x$: $2\sin^2 x = 1 - \cos(2x)$, so

$$\sin^2 x = \frac{1 - \cos(2x)}{2} = \boxed{-\frac{1}{2}\cos(2x) + \frac{1}{2}}.$$

This is $A\cos(Bx) + D$ with $A = -1/2$, $B = 2$, $D = 1/2$:

$$\text{period} = \frac{2\pi}{2} = \boxed{\pi}, \qquad \text{midline } \boxed{y = \tfrac{1}{2}}, \qquad \text{range} = [\tfrac{1}{2} - \tfrac{1}{2},\ \tfrac{1}{2} + \tfrac{1}{2}] = \boxed{[0, 1]},$$

with amplitude $1/2$. The period π is fundamental: if $p > 0$ is a period, then $\sin^2 p = \sin^2 0 = 0$, so p is a multiple of π and $p \geq \pi$.

The point rule $(u, v) \mapsto (u/2,\ \frac{1}{2} - \frac{1}{2}v)$ moves the parent cosine points $(0,1), (\pi/2, 0), (\pi, -1), (3\pi/2, 0), (2\pi, 1)$ to

$$(0,0), \quad \left(\tfrac{\pi}{4}, \tfrac{1}{2}\right), \quad \left(\tfrac{\pi}{2}, 1\right), \quad \left(\tfrac{3\pi}{4}, \tfrac{1}{2}\right), \quad (\pi, 0).$$

Because $A < 0$, the cycle starts at its minimum 0. Check directly: $\sin^2 \frac{\pi}{4} = \frac{1}{2}$, $\sin^2 \frac{\pi}{2} = 1$, $\sin^2 \frac{3\pi}{4} = \frac{1}{2}$, $\sin^2 \pi = 0$.

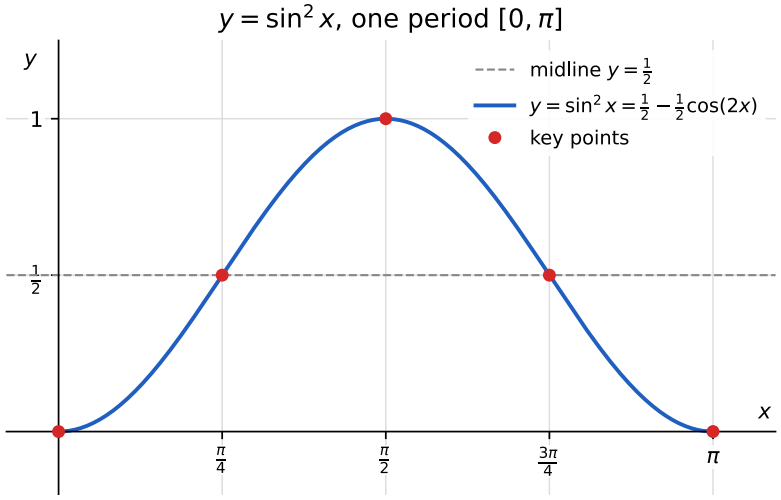


Figure 4: One period of y equals sin squared x on 0 to pi, which equals one half minus one half cos 2x, with midline y equals one half, minimum 0 at the ends and maximum 1 at pi over 2